TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Standardization of Hydrochloric Acid with Standard Sodium Carbonate Solution

The same acid certified a second, independent way — directly against a primary standard base, with the two-stage neutralisation of carbonate and the methyl-orange endpoint.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 15 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to standardise the HCl solution directly against a **primary standard** — sodium carbonate, weighed exactly (anhydrous Na_2CO_3, dried, is a primary standard for acids). Carbonate neutralises in **two stages**:

$$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl \quad \text{(stage 1, phenolphthalein endpoint)}$$
$$NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2 \quad \text{(stage 2, methyl orange endpoint)}$$

With **methyl orange** the titration runs through both stages to the full neutralisation:

$$Na_2CO_3 + 2\,HCl \rightarrow 2\,NaCl + H_2O + CO_2 \quad \text{(1 mol carbonate : 2 mol acid)}$$

2. Procedure

1. Prepare the standard Na_2CO_3 solution (0.100 M: 2.65 g per 250 mL, Experiment 11) from freshly dried anhydrous salt.
2. Pipette 25.0 mL of the carbonate into a conical flask; add 2–3 drops of methyl orange (solution yellow).
3. Fill the burette with the HCl; titrate with swirling. Near the endpoint, the CO_2 being released blurs the change — the professional finish is to **boil the solution briefly** near the endpoint to expel CO_2, cool, and complete the titration to the first permanent orange-pink tinge.
4. Repeat to concordance and calculate with the 1 : 2 ratio.

3. Worked calculation (model data)

STANDARDISATION RESULT

Standard Na_2CO_3: 0.100 M. Volume pipetted: 25.0 mL. Concordant titre of HCl: 49.2 mL.
Moles of Na_2CO_3 = $0.100 \times 0.0250 = 2.50 \times 10^{-3}$ mol.
Moles of HCl = $2 \times 2.50 \times 10^{-3} = 5.00 \times 10^{-3}$ mol (both stages counted by methyl orange).
Molarity of HCl = $5.00 \times 10^{-3} / 0.0492$ = **0.1016 M**.
Equivalents cross-check: 0.100 M Na_2CO_3 = 0.200 N; N(HCl) = $0.200 \times 25.0/49.2 = 0.1016$ N = 0.1016 M ✓.

Reconciling with Experiment 14: if the two standardisations of the "same" acid disagree beyond titration error, something is wrong in one of them — and suspicion falls first on the two-link chain of Experiments 13–14, because the direct primary-standard route of this experiment has fewer places to hide an error. Independent routes to one number, agreeing, are how laboratories trust a bottle.

4. The two-stage story, for the viva

With phenolphthalein the same titration would stop at stage 1 (half-neutralisation, carbonate → hydrogencarbonate) at exactly **half** the methyl-orange titre. That single flask thus demonstrates both of Topic 3's indicator lessons at once: the indicator chooses which equivalence the eye can see, and the stoichiometry follows the indicator. The boiling step exists because dissolved CO_2 (carbonic acid) makes the methyl-orange endpoint drift early and fade — removing it sharpens the change to a single drop.

5. Precautions

- The anhydrous Na_2CO_3 must be dried (about 270 °C, then desiccator-cooled) and weighed promptly — it is hygroscopic enough to matter at this level.
- Boil off CO_2 near the endpoint as directed; a "permanent" colour that fades in 30 seconds is CO_2 redissolving, not overshoot.
- Do not confuse the two titres: phenolphthalein titre = half the methyl-orange titre. Using the wrong factor doubles or halves the reported molarity.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Write both stages of the carbonate neutralisation and state which indicator catches which endpoint.
2. 25.0 mL of 0.100 M Na_2CO_3 needs 48.6 mL of HCl (methyl orange). Find the molarity.
3. In the titration of Q2, what would the phenolphthalein titre be, and what reaction ends there?

4. Why is the solution boiled near the methyl-orange endpoint?
5. Why is dried anhydrous Na_2CO_3 a primary standard while NaOH is not?
6. Experiment 14 gave the acid as 0.0971 M; this experiment's model gives 0.1016 M. What does a real disagreement of this size demand, and what is the first suspect?

Answer key

1. Stage 1: $Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$ (phenolphthalein endpoint, pink → colourless). Stage 2: $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$ (methyl orange endpoint) — total 2 mol HCl per mol carbonate.
2. Moles carbonate = 2.50×10^{-3}; moles HCl = 5.00×10^{-3}; M = $5.00 \times 10^{-3}/0.0486 = 0.1029$ M.
3. Half of 48.6 = 24.3 mL; the reaction ends at hydrogencarbonate (stage 1 only).
4. Neutralisation releases CO_2, which dissolves as weak carbonic acid and makes methyl orange change early and then drift back. Boiling expels the CO_2, so the final colour change marks the true endpoint sharply and permanently.
5. Dried Na_2CO_3 is obtainable pure, has a definite composition, and does not change in air during weighing; its weighed mass is a known amount of substance. NaOH absorbs water and CO_2 continuously, so its mass never equals a known amount of alkali — it can only be certified by titration.
6. It demands repetition of both standardisations — a disagreement of this size (about 4%) is far outside titration error, so at least one value is wrong. First suspect: the secondary-standard chain of Experiments 13–14. One clean mechanism matches the direction seen: if the phenolphthalein endpoint was overshot in Experiment 13, the titre there is too large, the NaOH gets certified too weak, and Experiment 14 — calculating with that too-weak value — reports the acid too low. The direct primary-standard route of this experiment is the one to trust while the chain is re-checked link by link.